

Heard but Not Studied: Mapping the Gap Between Health Research and Women's Lived Experiences

Data 510 Capstone

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I. Abstract

Women represent just over half the global population, yet for decades they have been systematically underrepresented in biomedical research. Our project aims to help close this gap by assessing the most common health issues raised in women's online discussion forums and the extent of research published on these issues. This brings us to our research question, "What women's health symptoms and conditions are most discussed on r/WomensHealth, and to what extent are these topics represented in clinical research on PubMed and ClinicalTrials.gov?" To answer this, we performed topic modeling and keyword frequency counts of posts that anonymous users have submitted to Reddit's r/WomensHealth forum, obtained the most commonly discussed topics and their keywords, and used them to pull research data from PubMed and ClinicalTrials.gov, which are both hosted by the National Institutes of Health, and analyzed the gaps between discussion count, paper count, and trial count. We found that topics affecting both genders are 3.85 times more researched than those affecting only women. We also found that topics that had an expected paper count that was over 10 times the actual paper counts were probiotics (for vaginal health), Bartholin cysts, female hormonal acne, unprotected sex, painful sex (Dyspareunia), interstitial cystitis, and bloating. These topics show the greatest disparity between discussion and research, prompting stakeholders to make data-driven decisions about which research to fund.

II. Introduction

Women's health is a crucial part of health research, directly addressing conditions that affect women. According to the World Health Organization, "Gender norms, power relations and structural inequalities influence women's exposure to health risks, access to care, quality of services and responsiveness of health systems. As a result, women may experience diseases differently from men, present with different symptoms and have distinct care needs at different stages of life (World Health Organization, n.d.)." Many factors contribute to women's health, but biological understanding and increased scientific research are critical to its development.

Historically, men's bodies were treated as the default in medical study design, with findings generalized to women despite important biological differences (Balch, 2024). Most of the time, this was not accidental; researchers often actively excluded female subjects due to concerns about hormonal variability interfering with results, the cost of housing female animals, and even due to policies designed to shield women of reproductive age from experimental drug exposure (Balch, 2024). The consequences have been serious, where women experience adverse drug reactions at twice the rate of men (Zucker & Prendergast, 2020), wait longer for both diagnosis and pain relief (Merone et al., 2022), and are more likely to be misdiagnosed during serious medical events (Merone et al., 2022). This is a pattern so well documented that it has been given a name, Yentl Syndrome, which is the phenomenon where a woman's illness must first mirror a man's presentation before it is taken seriously (Merone et al., 2022). Conditions which affect women, including endometriosis, uterine fibroids, autoimmune disorders, and

Polycystic Ovarian Syndrome (PCOS), remain underfunded and understudied relative to their prevalence (Edwards et al., 2023). Endometriosis is estimated to affect around 10% of women worldwide, yet diagnosis can take up to 10 years, in part because endometriosis is frequently dismissed as psychological (De Corte et al., 2024).

Online health communities have emerged as a large source of what burdens women experience in healthcare. Platforms like Reddit host active communities for conditions such as r/PCOS, r/endo, r/Fibrosis, r/periods, r/vaginismus, r/WomensHealth, and so much more. Women can share symptoms, treatments, healthcare experiences, and most importantly, gain knowledge. These discussions are extremely important, as they reflect the lived experiences of women that are often missing from published clinical research. Treating these spaces as data offers a way to locate that gap, or “missingness”, between what women experience versus what the research community prioritizes. To further investigate this gap, we analyzed established research and clinical trials, specifically in PubMed and ClinicalTrials.gov, alongside what women frequently discuss on Reddit, to identify the most underrepresented areas of women's health research.

Research Question: What women’s health symptoms and conditions are most discussed on r/WomensHealth, and to what extent are these topics represented in clinical research on PubMed and [ClinicalTrials.gov](https://clinicaltrials.gov)?

III. Impact and Stakeholders

Women have been historically underrepresented and underprioritized within medical and clinical published research, leading to a negative impact on women’s ability to understand their own health issues, receive diagnosis, treatment, and easily navigate healthcare systems. If the gender gap disparity in published health research is not closed, health systems will continue to burden women with self-advocacy in spaces that have not been designed to support them. Women will remain understudied, underfunded, and misunderstood. There will be a fully mapped understanding of what women discuss online, where published health research is missing, which topics are understudied and which are not studied at all, and what should be prioritized in future published research.

The primary stakeholder for this project is women. Stakeholders who will make that happen are medical researchers, clinical trial designers, and public health policymakers who decide where to allocate research funding and direct study enrollment. This project produces a data-driven, ranked map of women's health conditions that are widely discussed in patient communities but underrepresented in clinical literature, giving decision-makers a concrete, evidence-based starting point for prioritizing understudied conditions. Secondary stakeholders include patient advocacy organizations (such as endometriosis or PCOS foundations) who can use gap findings to strengthen funding arguments, and then women patients themselves, who stand to benefit downstream if research priorities shift toward conditions they disproportionately experience.

IV. Background & Related Work

Women's ability to be heard, listened to, researched, and cared for in our society has always been a struggle. Legislative intervention has tried to correct this imbalance. In 1993, the U.S. Congress required the inclusion of women in federally funded clinical trials (Balch, 2024), and the 2016 Twenty-First Century Cures Act went further by mandating that trial results be stratified by sex, gender, race, and ethnicity (Balch, 2024). Despite this progress, gaps remain substantial. Studies have been done to evaluate the National Institutes of Health (NIH) funding patterns. These studies have exposed the persistent disparities favoring diseases that predominantly affect men. As of May 12, 2026, PMOS (Polyendocrine Metabolic Ovarian Syndrome) is the current term for PCOS to more accurately reflect the condition and its presentation; however, since most studies and communities still refer to it as PCOS, we will use PCOS for consistency.

There have been many studies that have made progress on quantifying the gap from participant data (Feldman et al., 2019). For example, a 2019 study quantified sex bias in clinical studies using automated data extraction to assess the magnitude of female underrepresentation. They found substantial underrepresentation of women in 7 of the 11 disease categories they analyzed (Feldman et al., 2019).

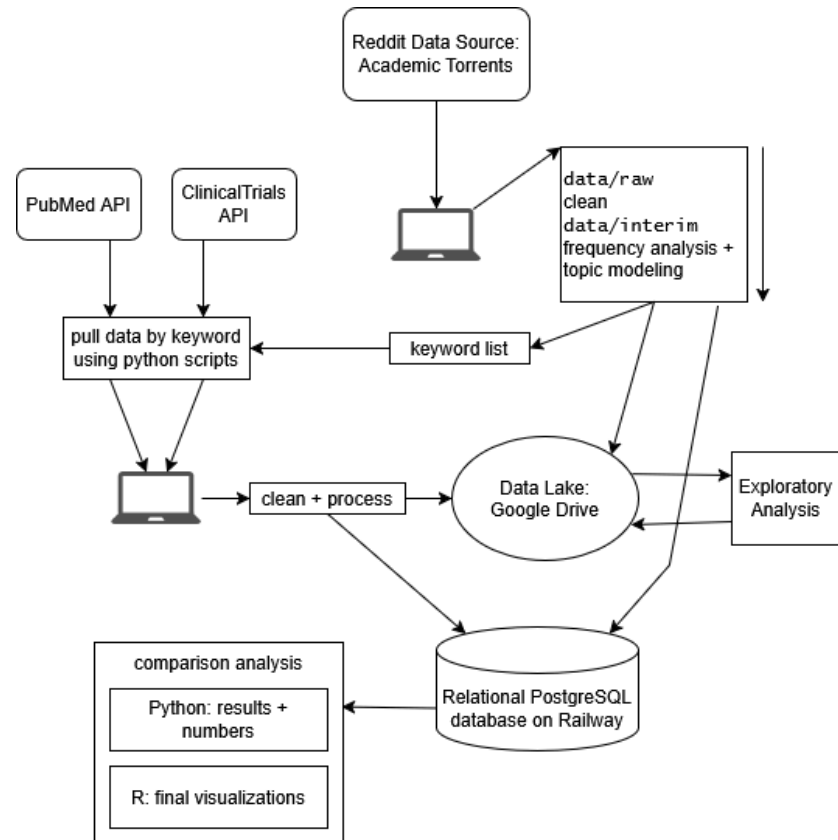
More recently, the World Economic Forum partnered with McKinsey Health Institute to create a Women's Health Impact Tracking platform to measure and close the women's health gap. This platform highlights and tracks statistics related to Clinical Trials published regarding women's health, mortality rates of conditions impacting women, and highlights indicator data for women's health. 1.<https://womens-health-impact-tracking-platform.weforum.org/>

There is new research on these kinds of spaces, including a 2025 study analyzing online communities formed around menopause (Funnell et al., 2025). Their main research angle was the quality of information in these communities, examined through thematic analysis. However, there is very little research comparing women's lived experiences with published health literature. Our project aims to help close this gap.

V. Data & Data Engineering

Our project has three data sources: Reddit data obtained via Academic Torrents (Watchful1, 2026); and PubMed and ClinicalTrials data obtained using the Python packages PaperScraper (Born, 2026) and PyTrials (Cavalcante, 2024), respectively. Below, we outline the project's data extraction and engineering. The figure below presents a flowchart of our data pipeline.

Engineering Pipeline



Git Repository

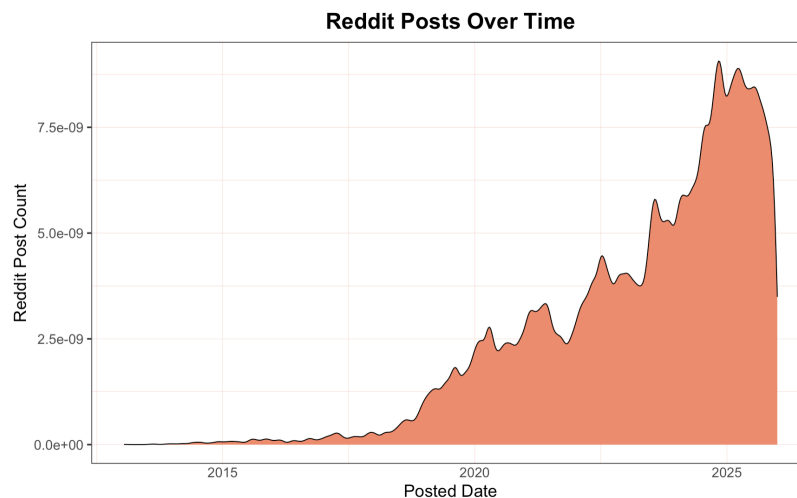
Our project is hosted on GitHub, where all of our work lies for reproducibility. This is our GitHub folder layout:

```
└─ GenderDisparityinResearch/
   │
   │ └─ data/
   │ │   └─ external/
   │ │   └─ interim/
   │ │   └─ processed/
   │ │   └─ raw/
   │ │
   │ └─ deliverables/
   │ │   └─ M1-proposal/
   │ │   └─ M2-data-summary/
   │ │   └─ M3-poster-draft/
   │ │   └─ M4-writeup-draft/
   │ │   └─ M5-final/
   │ │
   │ └─ notebooks/
   │ │   └─ Article_Data_APIs/
```

```
├── Reddit_Data/
│   ├── src/
│   └── studio/
│       ├── briefs/
│       └── critiques/
```

Reddit Data

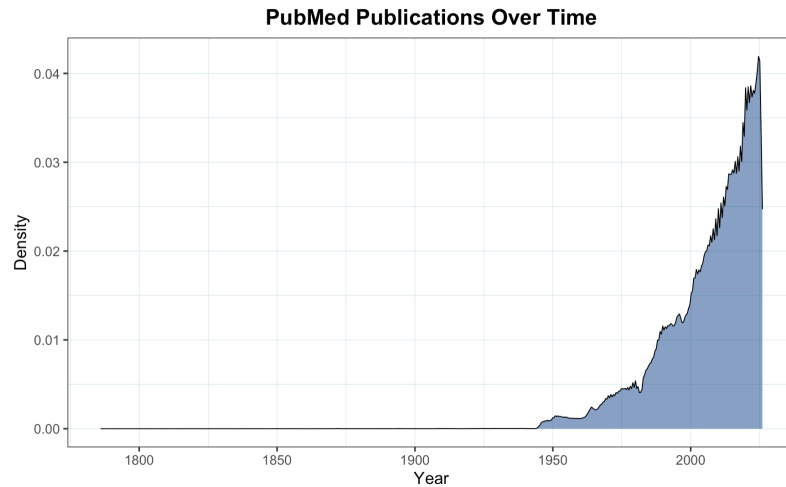
The source contained all Reddit user posts and comments from 2005-06 to 2025-12 in the top 40k subreddits. We downloaded its torrent file, opened it with qBittorrent, selected the r/WomensHealth subreddit, and ran the torrent, which downloaded the subreddit's entire history of posts and comments. The raw data lives in `data/raw/subreddits25/` and has both `WomensHealth_comments.zst` (98MB) and `WomensHealth_submissions.zst` (43MB). We unpacked the submissions .zst file and put the contents in `data/interim/subreddits/WomensHealth/`, which is only a JSON file named `submissions` (370MB). All data saved locally was also uploaded to Google Drive, which served as our data lake for this project. We cleaned the data by removing null columns, null rows, and rows with deleted or non-English content. We conducted frequency analysis and used BERTopic to identify the most commonly discussed topics and compiled a list of keywords based on our results.



PubMed Data

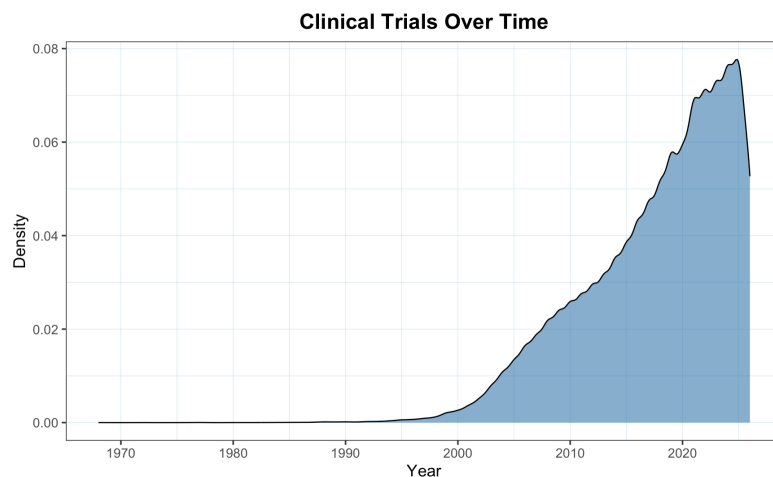
Data is retrieved from PubMed's API using the Python package PaperScraper, an open-source research software project that provides practical tools for scholarly metadata collection, publication search, full-text retrieval, citation analysis, and more. After obtaining the list of keywords, we wrote a Python script that iteratively queried for papers containing any of the keywords across different time blocks, spanning from 1786 to 2026. These separate blocks were saved as Pandas DataFrames, compiled into a single DataFrame using `pd.concat`, saved

as a CSV in `data/raw/` as `raw_pubmed_data.csv`, and uploaded to Google Drive. All of this work was done in `notebooks/paperscraper.ipynb`.



ClinicalTrials Data

The ClinicalTrials data was obtained in a similar fashion as the PubMed data. A Python package, PyTrials, was used to draw data using ClinicalTrials.gov's API. A script was made to pull data over iterations of different blocks of time and saved each as a Pandas DataFrame. The ClinicalTrials data spans the years 1968 to 2026. These separate DataFrames were merged using `pd.concat`, saved as a CSV in `data/raw/` as `raw_clinical_trials.csv`. All of this work was done in `notebooks/pytrials.ipynb`.

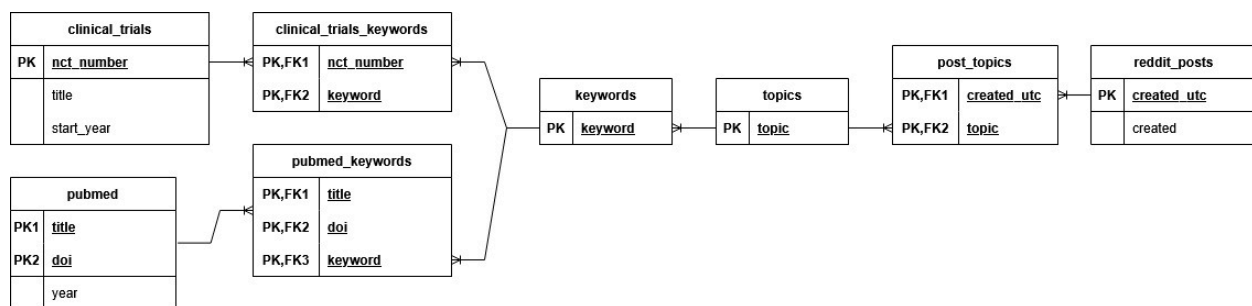


Google Drive

We used Google Drive as our data lake, a place to dump large amounts of data without incurring storage costs. It also serves as a place to collaborate on document files. In our Drive, we have four folders: **data/**, where we keep our large data files; **ERD/**, where we keep .drawio files of ERD sketches; and **notebooks/**, where we keep notebooks that utilize Google Colab's computing resources. Document files live in the home directory. All notebooks and anything necessary for reproducibility are copied to GitHub.

Relational Database

We have a PostgreSQL relational database hosted on Railway for a small monthly fee. This is where we link together the relationships between data, keywords, and topics. Arranging the data this way allows us to query all clinical trials, PubMed papers, and Reddit posts linked to a topic. This database includes a view that aggregates the post, clinical trial, and paper counts per topic for easy export to a Pandas DataFrame for comparative analysis. The figure shown below presents the ERD for our relational database.



VI. Ethics & Responsible Use

As this project works with health data, ethical considerations are imperative. The data needed for this analysis is extremely personal, and posts can be linked to a person in the Reddit data. This is why the username column is dropped entirely during data cleaning. Within the analysis, no personal information from the post title, comments, or post contents was used. Overall, this project wants to acknowledge that Reddit data contains valuable experiential knowledge that is being used for a good cause. It is acknowledged that using this type of data incorrectly can be misleading and divert attention from important topics.

PubMed publications data, as well as Clinical Trials data, were collected using Python scraper libraries with valid licenses and open-source licenses. Data pulled from PubMed included study title, abstract, DOI, journal, authors, and year. No personal information was abstracted from any columns pulled, and only raw counts were used in the final analysis dataframe. Data pulled from [ClinicalTrials.gov](https://clinicaltrials.gov) included NCT Number, primary outcome

measures, study title, sponsor, collaborators, sex, age, enrollment, summary, funder type, study type, start date, and completion date. No personal information was abstracted from any columns pulled, and only raw counts were used in the final analysis dataframe.

This study is also limited by the data, which affects how the results can be interpreted. PubMed and [ClinicalTrials.gov](https://clinicaltrials.gov) were the only publication data pulled. This does not capture or include research in other studies, or smaller publishers. Women's experience data was pulled from Reddit, which has a very restricted audience, mostly individuals in the United States, individuals with access to technology, as well as knowledge of this community.

Finally, it is extremely important to note that this project is looking at disparities in women's health in research without taking into account the intersectionality of race and gender. Women of color face systemic components, including stereotyping, insurance access, and accessible healthcare, that all contribute to the imbalance of equity for racial groups within healthcare. Medical literature overall struggles to report data regarding sex and gender; it struggles even more so with reporting race and ethnicity (Bierer et al., 2022). As well as the intersection of race and gender, this project is also not taking into account individuals outside of the gender binary that still might be affected by issues regarding the female reproductive system.

VII. Methods & Evaluation

Frequency Analysis

Frequency analysis is a computationally light way to identify the most commonly mentioned keywords, which is useful for assessing the topic makeup of r/WomensHealth at face value, assisting with data cleaning, and leading into more advanced topic modeling steps. We removed stopwords, common words that provide no insight into what the text is about ("and", "this", "of"), before creating frequency counts of keywords and bigrams (two-word keywords). To assess whether a count is statistically meaningful, a log-likelihood ratio/G-test can be done instead of a Chi-squared test, as Chi-squared tests assume normal distribution when word counts are not normally distributed. For example, a study showed that words with a frequency of less than 1 in 50,000 make up about 20-30% of typical English news reports (Dunning, 1993). This means the distribution has longer tails than a normal distribution. Performing a G-test using log-likelihood is more accurate than a Chi-squared test, according to Dunning (1993). To do this on our dataset, we compute the frequency of our top keywords and compare it to their expected frequencies in English using the Python package wordfreq (Speer, 2022), to obtain a Bonferroni-corrected p-value using the G-test. The top 50 keywords had p-values near zero, indicating that they occur more frequently in r/WomensHealth than in English.

BERTopic

A pre-loaded model that finds topic clusters, identifies keywords associated with a topic, and provides example text entries to assist the exploration and analysis of topics most discussed in the Reddit data.

Evaluation: The following tests & expected good result:

- C_V: 0–1; higher is better (often 0.5–0.7+ after good preprocessing).
- UMass: ≤ 0 ; closer to 0 is better (very negative = poor co-occurrence).
- Silhouette: $[-1, 1]$; > 0.2 modest, > 0.5 strong (text often yields modest values).
- Diversity: $[0, 1]$; > 0.85 suggests distinct vocabularies, but check for over-fragmentation.
- Entropy (norm): $[0, 1]$; closer to 1 = balanced topic sizes; closer to 0 = dominance by few topics.

Topic Mapping

As each study from both PubMed and Clinical Trials was retrieved via their APIs, a keyword was assigned to each instance. However, if that keyword matched more than once to a study, that study or trial was repeated in the final datasets. This is true for the Reddit data as well. As a Reddit post was assigned a keyword, multiple keywords were assigned to one post. Keywords pulled are as follows: menstrual cycle, irregular menstruation",menorrhagia, heavy menstrual bleeding, abnormal uterine bleeding, dysmenorrhea, period cramps, amenorrhea, breakthrough bleeding, intermenstrual bleeding, combined oral contraceptives, progestin-only pill, IUD, intrauterine device, copper IUD, levonorgestrel IUD, contraceptive side effects, birth control and mood, birth control and libido, tubal ligation, vulvovaginal candidiasis, recurrent yeast infection, bacterial vaginosis, urinary tract infection, UTI, boric acid vaginal suppository, vaginal pH, vaginal microbiome, probiotics vaginal health, pelvic pain, ovarian cyst, ovarian torsion, endometriosis, adenomyosis, pelvic floor dysfunction, pelvic floor physical therapy, vulvodynia, interstitial cystitis, Bartholin cyst, breast lump, fibroadenoma, breast pain, mastalgia, breast cancer screening, mammography, breast ultrasound, nipple pain, PCOS, polycystic ovary syndrome, hormonal acne, hirsutism, thyroid dysfunction, hypothyroidism, Hashimoto's thyroiditis, perimenopause, menopause hormone therapy, HRT, hot flashes, night sweats, sexually transmitted infection, STI, painful intercourse, dyspareunia, libido, sexual desire, unprotected sex, pregnancy test, medical abortion, misoprostol, mifepristone, abortion access, hair loss, alopecia, iron deficiency anemia, ferritin, bloating, hemorrhoids, nausea, fatigue, sleep disturbance, heart palpitation, headache, migraine, and allergic.

To mitigate keyword repetition across studies, topic mapping was used to consolidate a single topic per study. Each keyword was manually grouped by similar attributes and themes. For example, all IUD keywords, intrauterine device, copper IUD, levonorgestrel IUD, were grouped in a single topic, IUD.

After mapping keywords to topics, each topic was ranked based on how it should be assigned to a study, trial, or post. For example, if a study was found to have UTI, period cramps,

and breast cancer, the topic hierarchy would assign that study to breast cancer. The final topics are as follows: bartholin_cyst, ovarian_torison, ovarian_cyst, endometriosis, adenomyosis, fibroadenoma, breast_cancer, interstitial_cystitis, tubal_litigation, abortion_policy, abortion, uti, bacterial_vaginosis, yeast_infection, sti_std, pcos, thyroid, menopause, vulvodynia, iud, implant_shot_ring_patch, oral_contraceptives, contraceptive_side_effects, pregnancy_test, boric_acid, probiotics, vaginal_ph_microbiome, feminine_products, period_tracking_apps, covid_vaccine, dysmenorrhea, amenorrhea, heavy_bleeding, abnormal_bleeding, irregular_periods, pms_pmdd, unprotected_sex, pelvic_floor, painful_sex, breast_pain, libido, hirsutism, hair_loss, iron_anemia, menstrual_cycle, hormonal_acne, bloating, nausea, fatigue_sleep, headache_migraine, allergic_reaction, heart_papliptations, hemorrhoids, hot_flashes_night_sweats.

Model Testing

The final data set used, topic_summary.csv, contains all 49 final topics, with their PubMed publication count, clinical trials count, and Reddit post count. Since the final analysis is count data, none below zero, Poisson regression was used. The AIC (Akaike Information Criterion) baseline was set to gauge the performance of the models tested. This metric helps compare models by considering how well the model fits the data and how complex it is (ScienceInsights, 2026). A null model was fitted to both the PubMed publication count and the clinical trial count.

The first models used publications and trials based on post count; glm() was used in R, family = "poisson". The AIC decreased for both models: for clinical trials, from 219700 to 214056, a decrease of 5644. PubMed decreased from 3486000 to 3215544, with a difference of 270,456. The post count coefficient was positive in both models and highly statistically significant. Based on this result, overdispersion was assessed to determine the next best steps for modeling this data. It was found that, across all counts, PubMed, Clinical Trials, and Reddit posts had extremely large variance relative to the mean. Poisson's distribution assumes the variance of the data to be equal to the mean, meaning the Poisson model needed to be changed to account for the overdispersion within the data. Quasi-Poisson was the next model tested. However, this model cannot produce an AIC value but does output a dispersion parameter for each model. This allows users to see how well overdispersion is being addressed. Once tested, the Quasi-Poisson model yielded a dispersion parameter of 8824.975 for clinical trials and 100929.1 for PubMed. This revealed a different type of model was needed to handle the overdispersion within the data. The next model tested was Negative Binomial Regression. Poisson only rescales the data's variance, not its shape. Negative Binomial Regression uses a different variance function in quadratic form to accommodate cases where the variance grows much faster than the mean (MetricGate, 2026). The AIC baseline for the null models was 812.9674 for clinical trials, and 1105.619 for PubMed. Negative Binomial Regression was tested for predictor, post count, and response, publication and trials. The AIC for clinical trials

decreased from 812.9674 to 812.46, a difference of 0.5074. The AIC for PubMed decreased from 1105.619 to 1102.4, a difference of 3.219. Although the drop was extremely small, the dispersion parameter was set to 1 for both models, indicating that Negative Binomial Regression is the best model for this data.

Baseline AIC Table

AIC	Model	Variable
219700	Poisson	Clinical Trials
3486000	Poisson	PubMed
812.9674	Negative Binomial	Clinical Trials
1105.619	Negative Binomial	PubMed

Adding a New Variable

This project's final model uses a variable that was manually added and decided on based on further research. Due to data limitations, extracting sex or gender information from studies and trials was nearly impossible. Based on a topic and the keyword within that topic, each was assigned a boolean value if this keyword would be applicable to both genders or only women. For example, the "libido" topic, which contains keywords "libido" and "sexual desire", was assigned TRUE, as it pertains to both women and men. The topic "probiotics," containing the keyword "probiotics vaginal health," was assigned FALSE because it applies only to women. Topics like "breast cancer screening", which do affect both men and women but are overtaken by research for women and affect women at a higher percentage, were assigned FALSE, only affecting women.

VIII. Results & Visualizations

We ran a negative binomial regression model to predict the PubMed paper count for a topic from its Reddit post count. The post count variable is log-transformed to account for a heavily skewed count distribution.

```
Call:
glm.nb(formula = paper_count ~ log(post_count), data = df,
init.theta = 0.5702602424,
link = log)
```

```

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)      8.3200     0.7618  10.922 < 2e-16 ***
log(post_count)  0.3540     0.1242   2.849  0.00438 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(0.5703) family taken to
be 1)

    Null deviance: 65.375  on 47  degrees of freedom
Residual deviance: 59.763  on 46  degrees of freedom
AIC: 1089.7

Number of Fisher Scoring iterations: 1

              Theta: 0.5703
            Std. Err.: 0.0973

2 x log-likelihood: -1083.6710

```

These results show statistical significance for a coefficient of 0.354. Since the predictor is log-transformed, for every 1% increase in posts, there is a 0.354% increase in paper count.

We organized topics into two groups: True if the topic affects both sexes; False if the topic affects only females. We added that as a predictor variable in our model.

```

Call:
glm.nb(formula = paper_count ~ log(post_count) + both_sexes,
       data = df, init.theta = 0.6949472419, link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)      7.3792     0.7662   9.631 < 2e-16 ***
log(post_count)  0.3852     0.1169   3.296 0.000981 ***
both_sexesTRUE   1.3493     0.3677   3.669 0.000243 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(0.6949) family taken to
be 1)

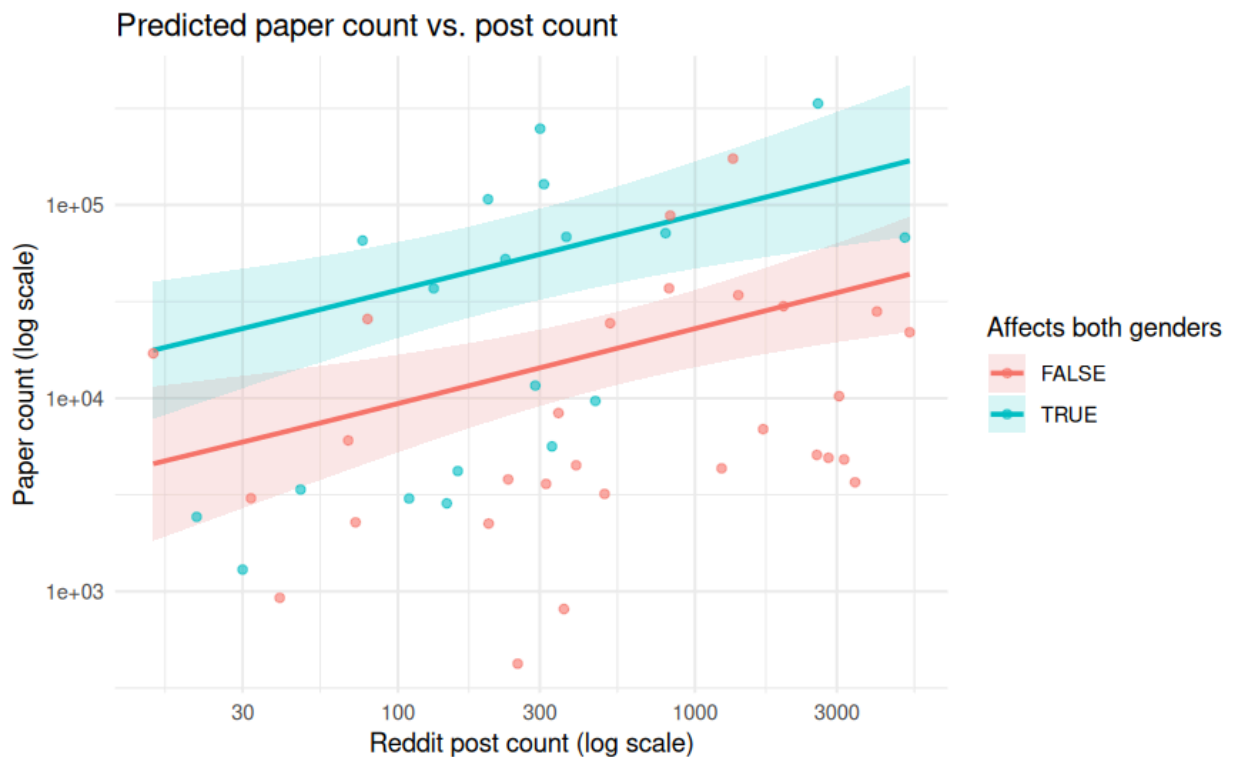
    Null deviance: 79.668  on 47  degrees of freedom
Residual deviance: 58.072  on 45  degrees of freedom
AIC: 1078.3

Number of Fisher Scoring iterations: 1

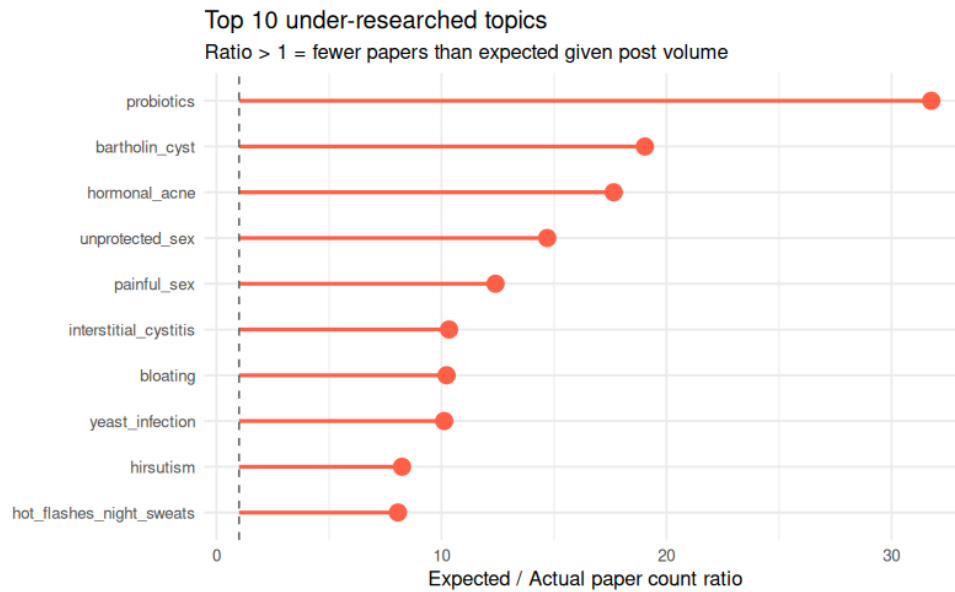
```

Theta: 0.695
Std. Err.: 0.121
2 x log-likelihood: -1070.333

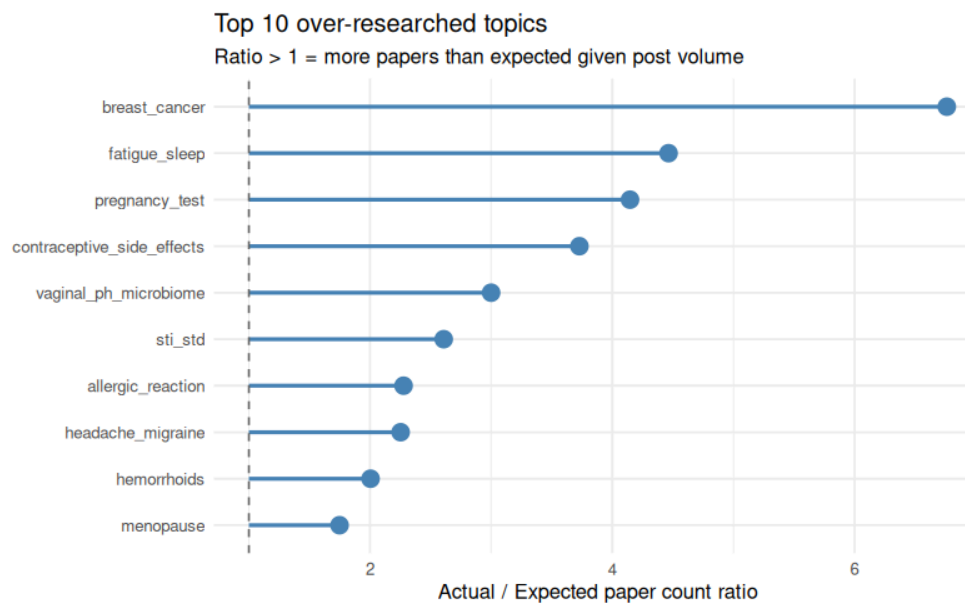
The coefficient of this model is very similar to the previous model, showing that a 1% increase in posts leads to a 0.3852% increase in papers. However, the coefficient for `both_sexesTRUE` tells us something entirely new: topics that affect both sexes have about x ($\exp(1.3493)$) times as many papers as topics that affect only females. The following chart shows the best-fit line for each sex category, where we can visually see the disparity.



Using the model to estimate an expected paper count based on `both_genders` and `post_count`, we can compare the actual paper count to the expected count by calculating ratios. The following chart shows the top under-researched topics with the largest expected-over-actual paper counts.



The most under-researched topic is vaginal health probiotics, with an expected-to-actual paper count ratio of 31.8, and Bartholin Cyst the second most under-researched topic, with a ratio of 19.0. The next chart below shows the relatively over-researched topics, with the largest actual-over-expected ratio.



The most-researched topic is breast cancer, with an actual-to-expected ratio of 6.7.

IX. Discussion, Limitations, & Future Work

Discussion

This project's results show that most underrepresented topics of women's health in research relative to what women talk about are: probiotics, bartholin cyst, hormonal acne, unprotected sex, painful sex, interstitial cystitis, bloating, yeast infection, hirsutism, and hot flashes or night sweats. The topics that are overrepresented compared to online discourse are: breast cancer, fatigue and sleep, pregnancy tests, contraceptive side effects, vaginal ph microbiome, STIs or STDs, allergic reaction, headache and migraines, hemorrhoids, and menopause. Alongside these findings, the analysis showed that topics affecting both sexes are researched about 3.82 times more often, confirming previous research and the known gap between research and women's health.

Limitations

As stated previously in VI. Ethics & Responsible Use: this project uses only two online publication websites and one online community (Reddit). The limitation is as follows: these results cannot be generalized to the full global scope of women's health research and to all women's experiences. The purpose of this study was to gain a deeper understanding of prior research on gender disparity in the medical field and to initiate a discussion in which women's voices are heard.

Another limitation pertains to how keywords are being used. Both PubMed and [ClinicalTrials.gov](https://clinicaltrials.gov) data sources were retrieved using keywords based on Reddit users; this alone is a limitation, as Reddit Users will not always use the vocabulary used in published research. As studies were pulled, keywords were collected if that word appeared in any text on the front API page. For example, if a Clinical Trial is about heart disease but uses postmenopausal women in the study, or menopause is a key variable, then "menopause" would be pulled as a keyword for the study. Some manual keyword checks against the studies were conducted to evaluate the pull's accuracy, but due to the dataset size, a full accuracy check was not possible.

Finally, referring back to VII. Methods & Evaluation, keyword mapping is another limitation of this project. Naturally, a study or trial will relate to many different words, and most of the time it is more complex than a single topic. The hierarchical mapping was implemented to mitigate this limitation, but with the available data, an accurate capture was not attainable.

Future Work

Future work should include a larger analysis with more data. This could be pulling data from more publication sources from other countries, smaller journals, etc. Regarding discussion data, surveys could be used to gain a more nuanced understanding of what women are talking about, what impacts them most, and what they struggle with, from a more diverse audience.

X. Conclusions & Recommendations

Earlier, we demonstrated how we calculated the ratio of expected-to-actual paper counts for each topic. After topic modeling of the Reddit forum r/WomensHealth, topics with significant discussion were selected to be included in our analysis, so they all have a real impact on women's lives. The topics with ratios of more than 10x include: probiotics (for vaginal health), Bartholin cysts, female hormonal acne, unprotected sex, painful sex (Dyspareunia), interstitial cystitis, and bloating. These topics show the greatest disparity between discussion and research, prompting stakeholders to make data-driven decisions about which research to fund.

Our research also showed that topics appearing in r/WomensHealth that impact both sexes receive about 3.85 times the research as topics affecting only the female sex. This calls for more research and funding to be allocated towards female health issues.

XI. Reproducibility & References

Reproducibility

All code is hosted on GitHub here:

<https://github.com/Addie-Shanti-Org/GenderDisparityinResearch>

A requirements.txt file is set up to ensure the proper environment settings. Our data is hosted on Kaggle here: [placeholder link] <https://www.kaggle.com/datasets>

References

Balch, B. (2024, March 26). *Why we know so little about women's health*. AAMC. Retrieved

July 26, 2026, from

<https://www.aamc.org/news/why-we-know-so-little-about-women-s-health>

Bierer, B. E., Meloney, L. G., Ahmed, H. R., & White, S. A. (2022). Advancing the inclusion of underrepresented women in clinical research. *Cell Reports Medicine*, 3(4), 5.

<https://doi.org/10.1016/j.xcrm.2022.100553>

Born, J. (2026, June 16). *PaperScraper*. Github.

<https://github.com/jannisborn/paperscraper/releases/tag/v1.0.0>

Cavalcante, J. (2024, May 1). *PyTrials*. PyPI. <https://pypi.org/project/pytrials/>

- De Corte, P., Klinghardt, M., von Stockum, S., & Heinemann, K. (2024). Time to Diagnose Endometriosis: Current Status, Challenges and Regional Characteristics—A Systematic Literature Review. *BJOG: An International Journal of Obstetrics & Gynaecology*, 132(2), 13. <https://doi.org/10.1111/1471-0528.17973>Digital Object Identifier (DOI)
- Dunning, T. (1993). Accurate Methods for the Statistics of Surprise and Coincidence. *ACL Anthology*, 19(1), 14. <https://aclanthology.org/J93-1003/>
- Edwards, T. L., Greene, C. A., Piekos, J. A., Hellwege, J. N., Hampton, G., Jasper, E. A., & Velez Edwards, D. R. (2023). Challenges and Opportunities for Data Science in Women's Health. *Annu Rev Biomed Data Sci.*, 6(23-45), 28. 10.1146/annurev-biodatasci-020722-105958
- Feldman, S., Ammar, W., & Lo, K. (2019). Quantifying Sex Bias in Clinical Studies at Scale With Automated Data Extraction. *JAMA Netw Open*, 2(7), 14. doi:10.1001/jamanetworkopen.2019.6700
- Funnell, E. L., McConnell, F., Martin-Key, N. A., Qian, L., Babbitt, K., & Bahn, S. (2025). Understanding Experiences of and Unmet Needs in Online Searches for Menopause Information: An Exploratory Survey. *JMIR Form Res*, 9(JISC), 16. 10.2196/75335
- Merone, L., Tsey, K., & Nagle, C. (2022). Sex Inequalities in Medical Research: A Systematic Scoping Review of the Literature. *Women's Health Report*, 3:1(49–59), 11. <https://doi.org/10.1089/whr.2021.0083>
- MetricGate. (2026, April). *Poisson vs Quasi-Poisson Regression*. MetricGate. <https://metricgate.com/blogs/poisson-vs-quasi-poisson-regression/>

ScienceInsights. (2026, March 9). *AIC and BIC in Regression: Formulas and Differences*.

ScienceInsights.

<https://scienceinsights.org/aic-and-bic-in-regression-formulas-and-differences/>

Speer, R. (2022). *wordfreq 3.1.1*. pypi.org. <https://pypi.org/project/wordfreq/>

UCLA Advanced Research Computing. (n.d.). *How do I interpret a regression model when some variables are log transformed*. Statistical Methods and Data Analytics.

<https://stats.oarc.ucla.edu/other/mult-pkg/faq/general/faqhow-do-i-interpret-a-regression-model-when-some-variables-are-log-transformed/>

Watchful1. (2026). *Subreddit comments/submissions 2005-06 to 2025-12*. Academic Torrents.

Retrieved June 16, 2026, from

<https://web.archive.org/web/20260724153410/https://academictorrents.com/details/3e3f64dee22dc304cdd2546254ca1f8e8ae542b4>

World Health Organization. (n.d.). *Women's Health*. World Health Organization Health Topics.

https://www.who.int/health-topics/women-s-health/#tab=tab_1

Zucker, I., & Prendergast, B. J. (2020, 6 5). Sex differences in pharmacokinetics predict adverse drug reactions in women. *Springer Nature Link*, 11(32), 14.

<https://doi.org/10.1186/s13293-020-00308-5>